

Lab 06:

HW 6

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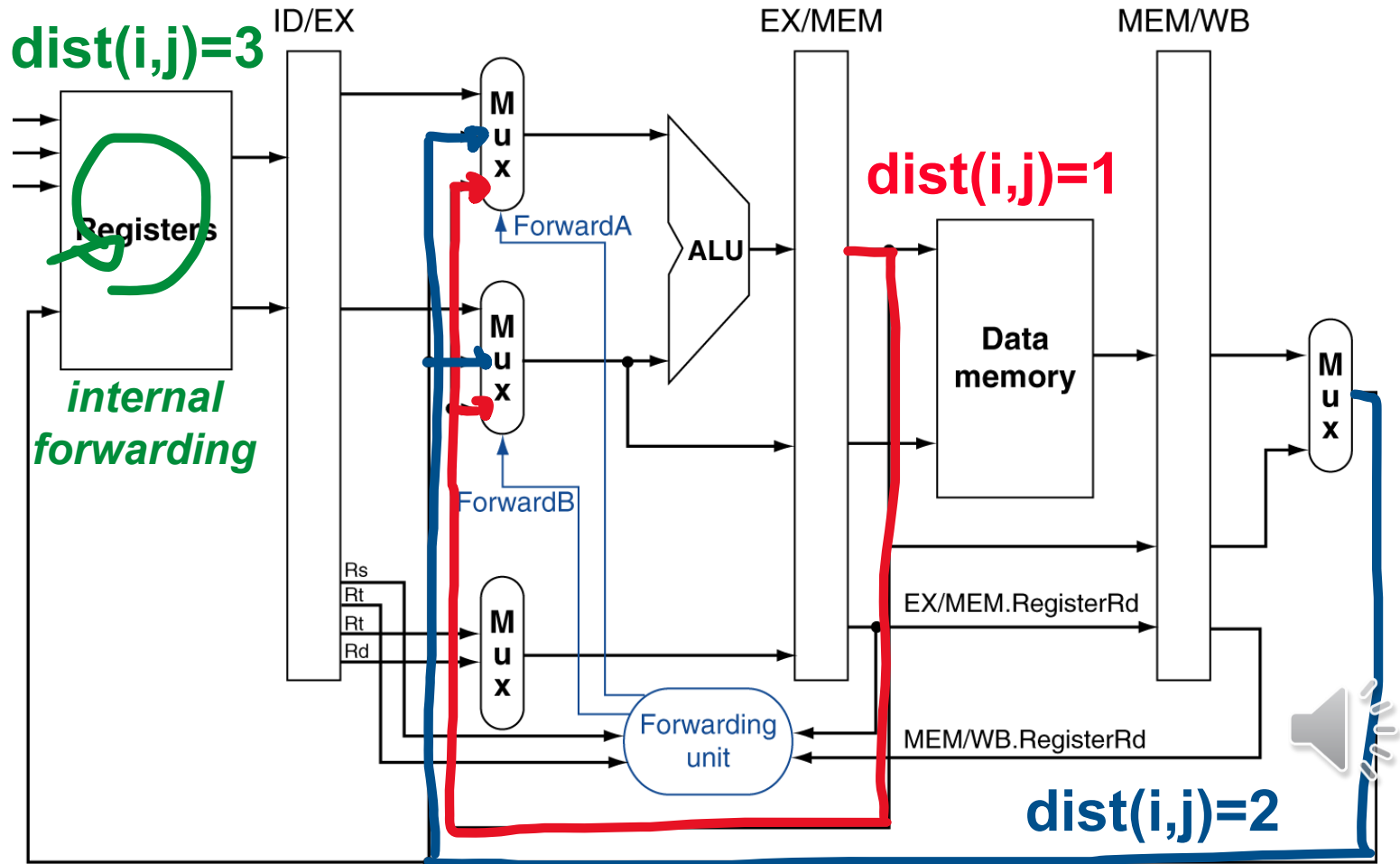
Overview

- ◆ You will implement additional features on your pipelined CPU
- ◆ What to do?
 - Implement a forwarding unit on your pipelined CPU
 - Enable various data forwarding to minimize pipeline stalls
 - ALU operands can be forwarded from prior ALU result
 - ALU operands can be forwarded from data memory
 - Register file supports internal data forwarding
 - Add branch prediction logic to facilitate complicated branch prediction
 - Predict the branch target + branch taken or not-taken



Data forwarding

- ◆ You must implement three different data forwarding methods



b. With forwarding

You still need stalls

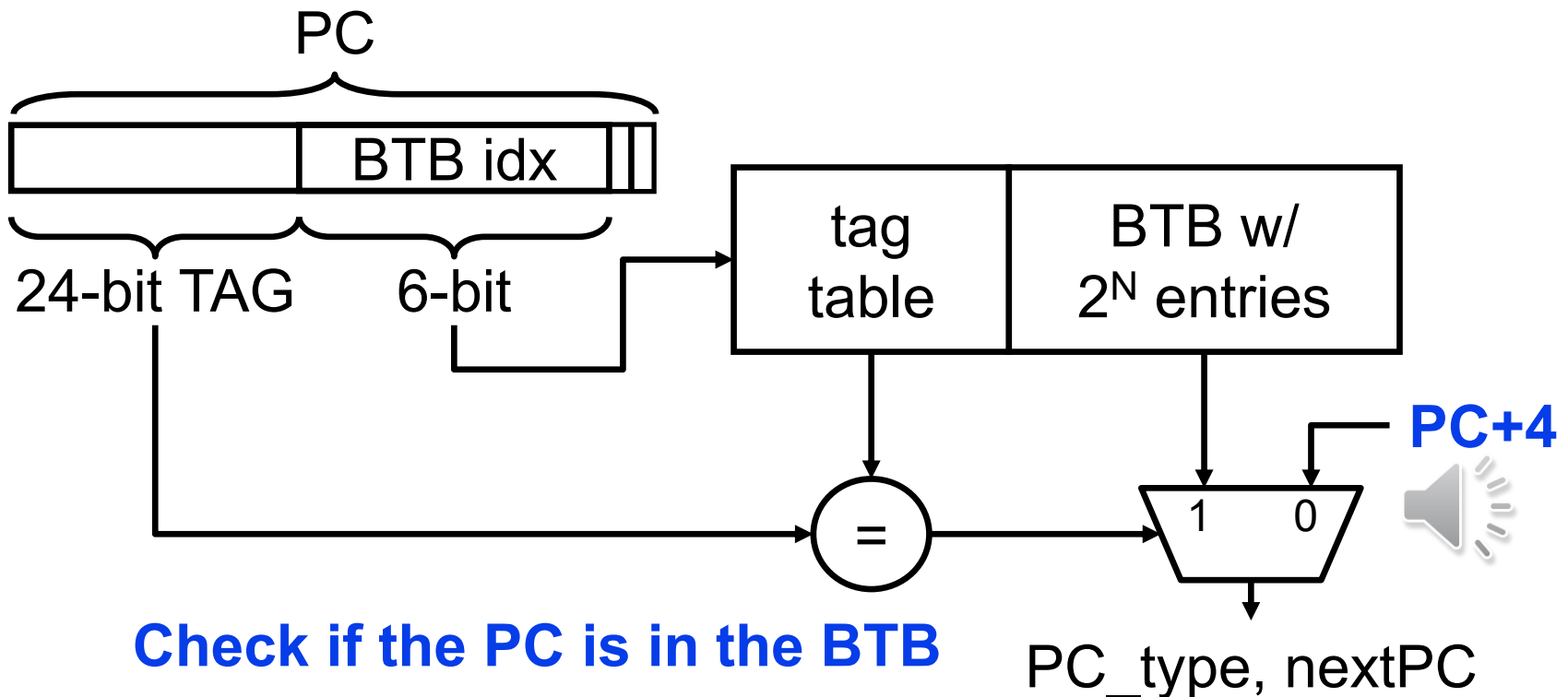
	R/I-Type	LW	SW	Br	J	Jr
IF						
ID						use
EX	use produce	use	use	use		
MEM		produce				
WB						

◆ You know the rules!



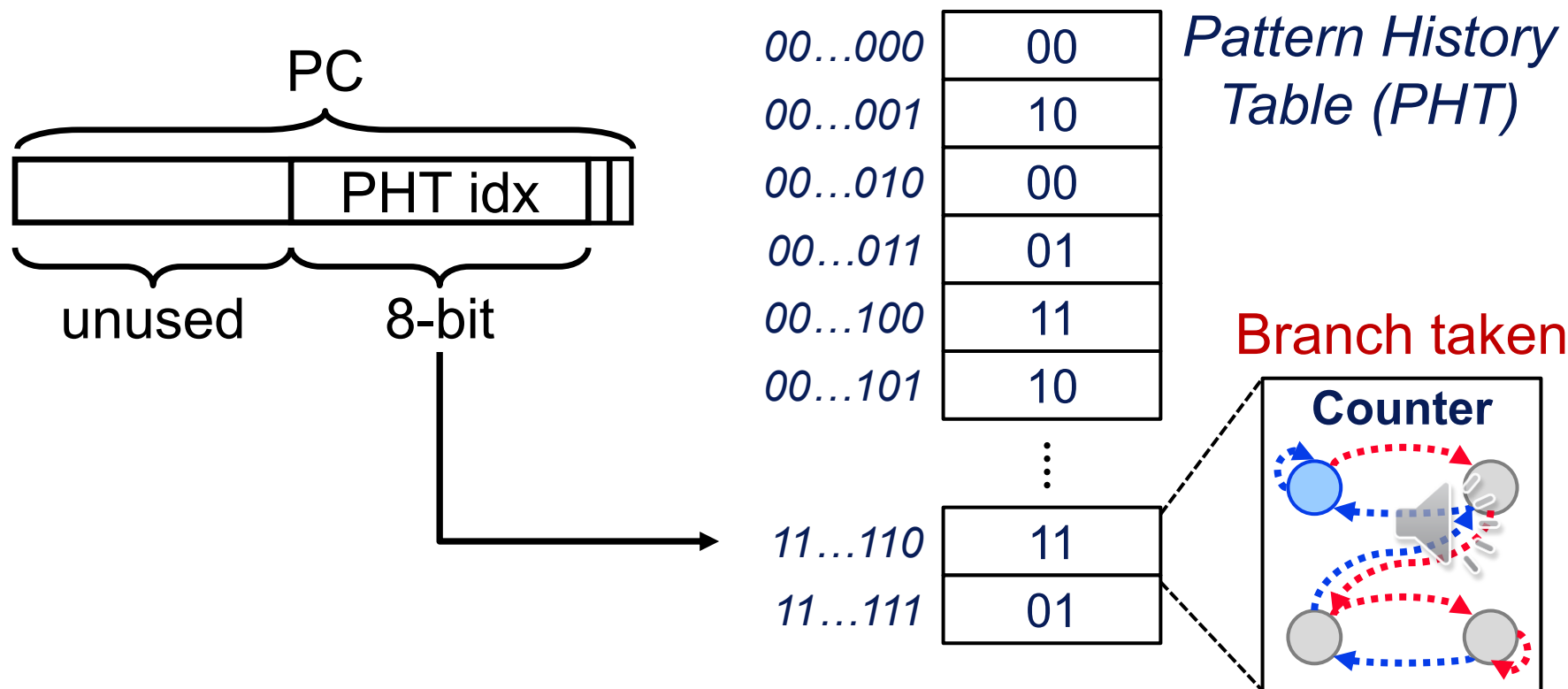
Branch Prediction - BTB

- ◆ You need to implement a BTB w/ 24-bit TAG + 6-bit idx (direct-mapped BTB)
- ◆ BTB stores two bit to indicate whether the BTB entry is empty (not initialized, jump, or branch)
- ◆ You don't need to support BP for JR-type instructions



Branch Prediction - Pattern

- ◆ Implement a 2-bit single-level branch predictor (**using saturation counters**)
- ◆ The PHT index should be 8-bit (allow multiple branches to share a PHT entry)



Updating BTB + PHT

- ◆ Update the branch predictor after the branch resolution
 - @ ID stage or EX stage depending on your implementation



Assignment

◆ Files:

- Add custom hardware modules to enable BP + Forwarding!
- Just make sure you modify Makefile if you add additional files



Submission

- ◆ Submission (Zip all the files)
 - For a two-people team: lab6_student_id1_student_id2.zip
 - For a one-person team: lab6_student_id.zip
 - You must follow the format (-10% for wrong file format)
 - Example:
 - lab6_2020102030.zip
 - lab6_2020102030_2022103040.zip
 - The zip file should contain:
 - every v files
 - lab6_report.pdf



Submission

- ◆ You need to write a 4+-page report
 - Explain the overall structure and how you implemented each program
 - Draw the hardware modules if needed ...
 - How did you implement the hazard detection unit, and stall logic ...
- ◆ Due: Mon. June 12th
 - No delay allowed (I need time to grade your project)

